MILITARY SPECIFICATION

TRANSMITTER, RATE GYROSCOPE TRU-2A/A

This amendment forms a part of MIL-T-27200C(USAF), dated 15 Jan 75, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Delete "first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications standards, and handbooks. The following specifications, standard, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Add SPECIFICATION, Federal and

"PPP-B-601	Boxes, Wood, Cleated Plywood"
"PPP-B-636	Boxes, Shipping, Fiberboard"

* Under SPECIFICATIONS, Military, MIL-C-5541 delete "and Aluminum Alloys."

* Under STANDARDS, Military, delete "MIL-STD-100" and add. "DOD-STD-100," same title.

Add boxed paragraph to the bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

* Under STANDARDS, Military, MIL-STD-704 delete "title" and add: new title "Aircraft Electric Power Characteristics." MIL-STD-810 delete "title" and add: new title "Environmental Test Methods & Engineering Guidelines." MIL-STD-882 delete "title" and add: new title "System Safety Program Requirements."

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* Add: "2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add. "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

* Add: "3.3.3.1 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

PAGE 6

3.6 Delete "MIL-STD-100" and add: "DOD-STD-100."

3.13.5 Delete existing paragraph and add:

" 3.13.5 Electrical transients. The transmitter shall be designed to withstand a spike of ±250V applied between pins A and B for 10 microseconds."

PAGE 9

Figure 1, upper right view Delete "1.88 ± .040 DIA" and add. "$\frac{1.20}{.84}$ radius." Also lower left view, delete "DIA"

PAGE 10

* Delete "4.1" and add new:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsbile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessry to assure supplies and services conform to prescribed requirements."

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* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective matrial."

4.3.2 Change "3.13.5" to read "3.13.4"

Delete "4.4.2 and add: new

"4.4.2 Test records. Throughout all qualificatior testing called for in paragraph 4.4 the contractor shall maintain complete test records and shall keep them available for inspection upon request by the local government representative. As a minimum the records shall include the following in accordance with MIL-STD-781: Operating sheet, log of failures, failure report, equipment repair sheet, failure tag and failed part summary analysis."

PAGE 15

4.6.1 Delete and add

"4.6.1 Examination of product. Prior to sealing, the gyroscope shall be inspected to determine compliance with 3.3, 3.4.3, 3.14, 3.15, and 3.16. This inspection need only be performed once on each unit produced. Upon final assembly, the gyroscope shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, weight and outline dimensions. The results shall be recorded. The weight and dimensions need only be recorded once for each unit."

PAGE 16

4.6.8.. Delete and add

"4.6.8.1 Electrical transients. The transmitter shall be subjected to an electrical spike with a 10 microsecond duration in both positive and negative polarities between pins A and B. The pulse shall be monitored and the test equipment adjusted so that 250 volts pulses are applied to the transmitter. After this test the transmitter shall meet the tests of 4.6.2 and 4.6.4."

4.6.11 Delete and add

"4.6.11 Sensitivity at low temperature. The transmitter shall be properly connected and placed in a chamber at a tempertaure of $-54^{\circ} \pm 2^{\circ}$ C ($-65^{\circ} \pm 4^{\circ}$F) for a period of 4 hours. At the end of the 4 hours and while still at low temperature, 28V dc shall be applied to the transmitter. After power has been applied for 10 minutes, the transmitter shall meet the test specified in 4.6.4, except the tolerance shall be ±10 percent or ±80 ua, whichever is greater. The test of 4.6.5 shall also be performed and passed, except the tolerance shall be +30, -25 percent. When at rest and at low temperature, subsequent to rotation, the transmitter shall meet the test specified in 4.6.3, except tolerance shall be ±70 ua."

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4.6.12 Delete and add:

4.6.12 Sensitivity at high temperature. The properly connected transmitter shall be subjected to a temperature of 71° ± 2°C (160° ± 4° F) for a period of 4 hours. While still at this temperature, 28V dc shall be applied. After 5 minutes of operation, the transmitter shall be subjected to and shall pass the test specified in 4.6.4, except the tolerance shall be ±10 percent or ±80 ua, whichever is greater. The test of 4.6.5 shall also be performed and passed, except the tolerance shall be ±20 percent. While still at high temperature, the transmitter shall pass the test specified in 4.6.3, except the tolerance shall be ±70 ua. The temperature shall be increased to 98°C (208°F) and held for 10 minutes. During this time, the transmitter shall be operating and shall indicate the imposed rate-of-turn within the specified limits."

PAGE 17

4.6.17 Line 2: Delete "Method 501" and add: "Method 501.1, Procedure II."

4.6.18 Line 2: Delete "Method 504, class 2" and add: "Method 504.1, category 6."

PAGE 18

4.6.19 Line 2: Delete "Method 503" and add: 503.1".

4.6.20 Delete existing paragraph and add

"4.6.20 Dynamic shock. The dynamic shock test shall be conducted in accordance with MIL-STD-810, Method 516.2, procedures I, II, and V. Except in procedure V, the lifted edge shall not be raised more than 2 inches above the horizontal bench top. Following the test specified in procedure I and after procedure V, the transmitter shall meet the individual tests."

4.6.21 Delete "Method 514" and add: "Method 516.2."

4.6.22 Delete second sentence.

4.6.23.1 Delete "Method 507" and add "Method 507.1."

4.6.23.2 Delete "Method 508" and add "Method 508.1."

4.6.23.3 Delete "Method 509" and add. "Method 509.1."

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4.6.23.4 Delete "Method 510" and add: "Method 510.1."

4.6.26 Delete "Method 515, procedure I, grade A" and add: "Method 515.2, procedure I, category A."

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4.6.28.1.2.e Change to read as follows:

"e. 4.6.11 and 4.6.12: Tolerance increases shall be as follows:

(1) 4.6.3 ± 80 ua

(2) 4.6.4 ± 20 percent

(3) 4.6.5 ± 35 percent"

* Delete "4.6.28.2 entirety and add: "4.6.28.2 Test Information. The contractor shall provide detailed test information as required by the contract.

* Delete the last sentence of paragraph 4.6.28.3.

Add the following new paragraph to Section 4.

4.7 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inpection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

* Delete Section 5 in its entirety and add

" 5. Packaging

5.1 Preservation-packaging: Preservation-packaging shall be level A, C, or Industrial as specified. (see 6.2).

5.1.1. Level A.

5.1.1.1. Cleaning. Transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2. Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3. Preservation application. Not applicable.

5.1.1.4. Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

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5.1.2. Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3. Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2. Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1. Level A. Transmitter packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2. Level B. Transmitter packaged as specified in 5.1.1.1. shall be packed in shipping containers conforming to PPP-B-636, class weaher-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1. apply.

5.2.3. Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3. Marking. In addition to any other markings required by the contract or order (See 6.2) interior and exterior containers shall be makred in accordance with MIL-STD-129."

* 6.4 Lines 9 and 10 delete "the Directorate of Airframe Engineering, Attn. ENFIF, Wright Patterson Air Force Base, Ohio 45433" and add "Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990."

* Delete "6.5."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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